LORAWAN Water Meter Upload and Control Protocol

1.Specific analysis of water meter upload protocol

68 10 78 56 34 12 00 00 00 81 1E 92 1F 06 2C 25 00 00 00 2C 25 00 00 00 35 70 01 00 00 00 18 00 11 43 00 11 05 19 20 40 00 25 16

68 isthe start of the frame 68H

10 is the meter type T

78 is the meter address A0

56 is the meter address A1

34 is the meter address A2

12 is the meter address A3

00 is the meter address A4

00 is the meter address A5

00 is the meter address A6 (A0、A1、A2、A3is the address of the water meter , from low to high)

81 As control character C

1E Is the data length field L(92 1F 06 2C 25 00 00 00 2C 25 00 00 00 35 70 01 00 00 00 18 00 11 43 00 11 05 19 20 40 00,It is 30 bytes in total)

92 Identify data DI_0

1F Identify data DI_1

06 is the serail NO SER

2C 25 00 00 00 Is the positive accumulated flow of 0.025m^3, 2C is the unit of flow m^3。

content	Data format	Unit	Byte number	description	备注
positive Accumulated flow	BCD	M^3	4	positive accumulated flow	example: Original data: 2C 25 00 00 00 Analysis method: 25 times 0.001 Result: 0.025

2C 25 00 00 00 Is the reverse accumulated flow 0.025 m^3, 2C is the unit of flow M3。

content	Data format	unit	Byte number	description	备注
Reverse Accumulated flow	BCD	M^3	4	Reverse Accumulated flow	example: Original data: 2C 25 00 00 00 Analysis method: 25 times 0.001 Result: 0.025

35 70 01 00 00 Is instantaneous flow rate 0.0170M3/h, 35 is the unit of instantaneous flow rate M3/h。

content	Data format	unit	Byte number	description	备注
Instantaneous flow	BCD	M3/h	4	Instantaneous flow	example: Original data: 35 70 01 00 00 Analysis method: 170 times 0.0001 Results: 0.0170M3/h

00 18 00 is temperature, unit ℃, 18.00℃

content	Data format	unit	Byte number	description	备注
temperature	BCD	℃	3	temperature	For example: Original data: 00 18 00 Analysis method: 1800 times 0.01 Result: 18.00

11 43 00 11 05 19 20 is the date May,11th,2019 00:43

content	Data format	unit	Byte number	description	备注
Current date	BCD	Year/ Month /day /hour /minute	7	Current date	For example: Original data: 11 43 00 11 05 19 20 Analysis method: 20190511004311 Result: on May 11, 2019 00: 43

40 00 is the status word battery voltage is low , The flow sensor erro Ec

content	Data format	unit	Byte number	description	备注
status word	HEX		2	status word Corresponding to Table 1 and Table 2	example: Original data: 40 00 Analysis method: HEX Low is in front and high is behind Result: 00 40

Table 3.1 Status ST The first byte defines table

	D0	D1	D2	D3	D4	D5	D6	D7
definition	Valve condition		Battery voltage	Water pipe burst Erro	Water pipe leakage Erro	Temperature sensor error	The flow sensor erro or the pipe empty	Water pipes installed positions erro
description	00:open 01:close 11:anomaly		0: nomaly 1: low voltage	0:nomaly 1: burst	0: nomaly 1: leakage	0:nomaly 1:Erro	0:nomaly 1:Erro	0:nomaly 1:reverse

Table3.2 Status ST The second byte defines table

	D0	D1	D2	D3	D4	D5	D6	D7
definition	Open valve mode	Close valve mode	External power	Open valve Erro	Close valve Erro	Forced Open valve	Forced Close valve	External power supply
description	0:unnormal open 1:normal open	0:unnormal close 1:normal close	0:unnormal 1:Low voltage	0:norma 1:erro	0:norm 1:erro	0:optional 1:Forced	0:optional 1:Forced	0: normal 1: Low voltage

25 check code CS(68 10 78 56 34 12 00 00 00 81 1E 92 1F 06 2C 25 00 00 00 2C 25 00 00 00 35 70 01 00 00 00 18 00 11 43 00 11 05 19 20 40 00summation)

16 is the end frame 16H

2. Downlink control protocol

Downstream set the upload cycle and so on through the AT command

The AT + SUPLDPA: 01440014 40,60,01440,00000 \ r \ n

Raw data feedback: AT+SUPLDPA:OK\r\nCommand:>

Upload interval :1440 minutes, upload delay :1440 minutes, upload delay seconds :60 seconds, active meter reading period :1440 minutes

Valve control instruction

3.1.7.1 Force open the valve

To send the message:

FE FE FE 68 10 78 56 34 12 00 00 00 04 04 A0 17 01 55 A1 16

68 is the frame starter 68H

10 is the meter type T (water meter)

78 is address A0

56 is address A3

34 is address A2

12 is address A3

00 is address A4

00 is address A5

00 is address A6 (A0, A1, A2, A3, A4, A5, A6 is the meter address 12345678, from low to high)

04 is the control code C (write data)

04 is the data length field L (A0 17 01 55) total 4 characters)

A0 is the data identifier DI0

17 is the data identifier DI1 (write valve control)

01 indicates the serial number SER

55 for open/close valve operation (open /close valve control operation)

A1 does the binary arithmetic sum for the check code CS (68 10 78 56 34 12 00 00 00 04 04 A0 17 01 55), excluding the overflow value exceeding FFH)

16 is the end 16H

To receive a message:

FE FE FE 68 10 78 56 34 12 00 00 00 84 05 A0 17 01 00 00 CD 16

68 is the frame starter 68H

10 is the meter type T (water meter)

78 is address A0

56 is address A3

34 is address A2

12 is address A3

00 is address A4

00 is address A5

00 is address A6 (A0, A1, A2, A3, A4, A5, A6 is the meter address 12345678, from low to high)

84 is control code C normal reply

05 indicates the data length field L (A0 17 01 00 00), a total of 5 characters.

A0 is the data identifier DI0

17 is the data identifier DI1 (write valve control)

01 is the serial number SER

00 00 is the system status (check the system status description in the "Read Metering Data" data message for details)

CD does the binary arithmetic sum for the check code CS (68 10 78 56 34 12 00 00 00 84 05 A0 17 01 00 00), excluding the overflow value exceeding FFH)

16 is the end 16H

3.1.7.2 Forcibly close the valve

To send the message:

FE FE FE 68 10 78 56 34 12 00 00 00 04 04 A0 17 01 99 E5 16

68 is the frame starter 68H

10 is the meter type T (water meter)

78 is address A0

56 is address A3

34 is address A2

12 is address A3

00 is address A4

00 is address A5

00 is address A6 (A0, A1, A2, A3, A4, A5, A6 is the meter address 12345678, from low to high)

04 is the control code C (write data)

04 is the data length field L (A0 17 01 99) total 4 characters)

A0 is the data identifier DI0

17 is the data identifier DI1 (write valve control)

01 is the serial number SER

55 for open/close valve operation (close valve control operation)

E5 adds binary arithmetic for check code CS (68 10 78 56 34 12 00 00 00 04 04 A0 17 01 99), excluding the overflow value exceeding FFH)

16 is the terminator 16H

To receive a message:

FE FE FE 68 10 78 56 34 12 00 00 00 84 05 A0 17 01 00 00 CD 16

68 is the frame starter 68H

10 is the meter type T (water meter)

78 is address A0

56 is address A3

34 is address A2

12 is address A3

00 is address A4

00 is address A5

00 is address A6 (A0, A1, A2, A3, A4, A5, A6 is the meter address 12345678, from low to high)

84 is control code C normal reply

05 is the data length field L (A0 17 01 00 00) 5 characters in total)

A0 is the data identifier DIO

17 is the data identifier DI1 (write valve control)

01 is the serial number SER

00 00 is the system status (check the system status description in the "Read Metering Data" data message for details)

CD does the binary arithmetic sum for the check code CS (68 10 78 56 34 12 00 00 00 84 05 A0 17 01 00 00), excluding the overflow value exceeding FFH)

16 is the end 16H

3.1.7.3 Force undo

To send a message:

FE FE FE 68 10 78 56 34 12 00 00 00 04 04 A0 17 01 FF 4C 16

68 is the frame starter 68H

10 is the meter type T (water meter)

78 is address A0

56 is address A3

34 is address A2

12 is address A3

00 is address A4

00 is address A5

00 is address A6 (A0, A1, A2, A3, A4, A5, A6 is the meter address 12345678, from low to high)

04 is the control code C (write data)

04 is the data length field L (A0 17 01 FF 4 characters in total)

A0 is the data identifier DI0

17 is the data identifier DI1 (write valve control)

01 is the serial number SER

FF for open/close valve operation (forced undo operation)

4C is the binary arithmetic sum for the check code CS (68 10 78 56 34 12 00 00 00 04 04 A0 17 01 FF), excluding the overflow value exceeding FFH).

16 is the end 16H

To receive a message:

FE FE FE 68 10 78 56 34 12 00 00 00 84 05 A0 17 01 00 00 CD 16

68 is the frame starter 68H

10 is the meter type T (water meter)

78 is address A0

56 is address A3

34 is address A2

12 is address A3

00 is address A4

00 is address A5

00 is address A6 (A0, A1, A2, A3, A4, A5, A6 is the meter address 12345678, from low to high)

84 is control code C normal reply

05 is the data length field L (A0 17 01 00 00 total 5 characters)

A0 is the data identifier DI0

DI1 for data identification (write valve control)

01 is the serial number SER

00 00 is the system status (check the system status description in the "Read Metering Data" data message for details)

CD does the binary arithmetic sum for the check code CS (68 10 78 56 34 12 00 00 00 84 05 A0 17 01 00 00), excluding the overflow value exceeding FFH)

16 is the end 16H

3.1.7.4 Open the valve freely

To send a message:

FE FE FE 68 10 78 56 34 12 00 00 00 04 04 A0 17 01 66 B2 16

68 is the frame starter 68H

10 is the meter type T (water meter)

78 is address A0

56 is address A3

34 is address A2

12 is address A3

00 is address A4

00 is address A5

00 is address A6 (A0, A1, A2, A3, A4, A5 and A6 are meter addresses 12345678, from low to high)

04 is the control code C (write data)

04 is the data length field L (A0 17 01 66 total 4 characters)

A0 is the data identifier DI0

17 is the data identifier DI1 (write valve control)

01 is the serial number SER

66 for open/close valve operation (free open valve)

B2 does the binary arithmetic sum for the check code CS (68 10 78 56 34 12 00 00 00 04 04 A0 17 01 66), excluding the overflow value exceeding FFH)

16 is the end 16H

To receive a message:

FE FE FE 68 10 78 56 34 12 00 00 00 84 05 A0 17 01 00 00 CD 16

68 is the frame starter 68H

10 is the meter type T (water meter)

78 is address A0

56 is address A3

34 is address A2

12 is address A3

00 is address A4

00 is address A5

00 is address A6 (A0, A1, A2, A3, A4, A5, A6 is the meter address 12345678, from low to high)

84 is control code C normal reply

05 is the data length field L (A0 17 01 00 00 total 5 characters)

A0 is the data identifier DI0

17 is the data identifier DI1 (write valve control)

01 is the serial number SER

00 00 is the system status (check the system status description in the "Read Metering Data" data message for details)

CD does the binary arithmetic sum for the check code CS (68 10 78 56 34 12 00 00 00 84 05 A0 17 01 00 00), excluding the overflow value exceeding FFH)

16 is the end 16H

3.1.7.5 Free closing valve

To send the message:

FE FE FE 68 10 78 56 34 12 00 00 00 04 04 A0 17 01 77 C3 16

68 is the frame starter 68H

10 is instrument type T (water meter)

78 is address A0

56 is address A3

34 is address A2

12 is address A3

00 is address A4

00 is address A5

00 is address A6 (A0, A1, A2, A3, A4, A5, A6 is the meter address 12345678, from low to high)

04 is the control code C (write data)

04 is the data length field L (A0 17 01 66 total 4 characters)

A0 is the data identifier DI0

17 is the data identifier DI1 (write valve control)

01 is the serial number SER

77 for open/close valve operation (free close valve)

C3 does binary arithmetic summations for check code CS (68 10 78 56 34 12 00 00 00 04 04 A0 17 01 77), excluding spillover values exceeding FFH)

16 is the terminator 16H

To receive a message:

FE FE FE 68 10 78 56 34 12 00 00 00 84 05 A0 17 01 00 00 CD 16

68 is the frame starter 68H

10 is the meter type T (water meter)

78 is address A0

56 is address A3

34 is address A2

12 is address A3

00 is address A4

00 is address A5

00 is address A6 (A0, A1, A2, A3, A4, A5, A6 is the meter address 12345678, from low to high)

84 is control code C normal reply

05 is the data length field L (A0 17 01 00 00 total 5 characters)

A0 is the data identifier DI0

17 is the data identifier DI1 (write valve control)

01 is the serial number SER

00 Indicates the system status. (For details, see System Status description in "Read Metering Data" data message.)

CD does the binary arithmetic sum for the check code CS (68 10 78 56 34 12 00 00 00 84 05 A0 17 01 00 00), excluding the overflow value exceeding FFH)

16 is the end 16H